

Aiman Siddiqui

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Software Engineer and MSc Computer Science student with experience building scalable web and mobile applications using React, Node.js, and TypeScript. Experienced in the full software development lifecycle from ideation and experimentation to production deployment and monitoring. Strong foundation in data structures, algorithms, object-oriented design, and complexity analysis. Passionate about AI-first engineering, leveraging AI-assisted tools for debugging, log analysis, rapid prototyping, and delivering high-quality code at speed.

Technical Skills

Programming Languages: JavaScript, TypeScript, Python

Frontend: React, React Native, HTML, CSS

Backend: Node.js, NestJS, FastAPI, REST APIs, GraphQL

Databases: PostgreSQL, MongoDB, MySQL, Redis

Cloud & DevOps: AWS (Lambda, S3, Cognito), Google Cloud Functions, Docker, Kubernetes, CI/CD (GitHub Actions, Jenkins)

Testing & Quality: Jest, Integration Testing, Automated Testing

Computer Science: Data Structures, Algorithms, Object-Oriented Design, Complexity Analysis, Problem Solving

AI & Data: TensorFlow, PyTorch, Scikit-learn, OpenCV, LLM Integrations

Development Practices: Agile, Cross-functional Collaboration, Code Reviews, Monitoring & Observability

Education

Saarland University

M.S. in Computer Science

Germany

Oct 2024 – Ongoing

Relevant Coursework: Machine Learning, Neural Networks, Cloud Computing

National University of Computer and Emerging Sciences

B.S. in Computer Science

Karachi, Pakistan

Jul 2017 – Jul 2021

Graduated with a **CGPA of 3.91**, awarded the **Bronze Medal** for exceptional academic performance.

Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, Database Systems

Experience

Research Assistant

DFKI

Jul 2025 – Present

Kaiserslautern, Germany

- Developed React frontend features on FastAPI (Python) backend services, designing and consuming REST APIs to deliver reliable, data-driven interfaces for research applications.
- Collaborated with researchers and engineers to translate complex research requirements into scalable and maintainable UI features.
- Integrated OAuth-based authentication and contributed to improving API performance, maintainability, and application reliability.
- Participated in iterative experimentation cycles to evaluate feature effectiveness and user interaction patterns.

Werkstudent Full Stack Engineer

WorkCentrix GmbH

Mar 2025 – Jun 2025

Saarbrücken, Germany

- Developed scalable web applications using Angular, PHP, and MySQL within an agile team, handling backend logic, database design, and frontend development.
- Built reusable Angular UI components that reduced frontend feature development time by 30%.
- Contributed to CI/CD pipelines and deployment workflows to support stable and reliable production releases.
- Collaborated with product and design stakeholders to refine requirements and deliver user-centric features.

Senior Software Engineer

Folio3 Pvt Ltd

Jan 2023 – Oct 2024

Karachi, Pakistan

- Led React.js and React Native development for Thrive, building maintainable reusable components with React Query to improve performance by 25% and support 10,000+ active users.

- Architected microservices for Genopalate using Node.js, NestJS, and AWS Lambda, orchestrating a distributed service-oriented pipeline across Lambda, webhook-driven third-party services, and callbacks, reducing API latency by 30%.
- Owned features end-to-end through GitHub Actions and Docker-based CI/CD pipelines, automated testing, deployment, monitoring, and production debugging.
- Created API documentation and engineering runbooks that reduced new-developer onboarding time by 40% and approximately 15 support tickets per week.
- Mentored junior engineers through code reviews, technical guidance, and collaborative problem solving.
- Developed AI-driven recommendation systems and LLM-powered chatbot features, increasing user engagement by 25%.

Software Engineer

Folio3 Pvt Ltd

Jul 2021 – Dec 2022

Karachi, Pakistan

- Implemented a real-time chatbot using WebSockets for Moodology and integrated third-party services including Shopify and Firebase across 10+ React Native applications serving 50,000+ active users.
- Engineered backend services for Clomr using Node.js APIs and Google Cloud Functions with asynchronous programming, improving processing efficiency by 30%.
- Developed and maintained full-stack applications using React, React Native, Node.js, and REST APIs across healthcare, e-commerce, and consumer-facing products.
- Integrated healthcare systems and APIs using EHR/EMR standards including FHIR and HL7.

Machine Learning Intern

Bytecorp

Jun 2020 – Jul 2020

Karachi, Pakistan

- Built and trained CNN-based image classification models using TensorFlow and applied transfer learning to improve model performance by up to 20% on limited labeled datasets.
- Optimized CNN architectures to reduce inference time by 25% while maintaining model performance.

Projects

ScanWise

AI-Powered Career Assistant

Next.js, TypeScript, shadcn/ui, FastAPI, PostgreSQL

- Built an AI-powered career assistant that analyzes resumes and job descriptions to generate compatibility scores, matched and missing skills, and personalized recommendations.
- Developed an interactive dashboard using Next.js, TypeScript, and shadcn/ui with the Next.js App Router to translate AI-generated analysis into clear, actionable insights.
- Engineered FastAPI and PostgreSQL backend services for document uploads, structured data extraction, and streaming AI responses.
- Used Claude Code and Cursor throughout development for rapid prototyping, debugging, and implementation.

Balcony PV Detection

M.Sc. Thesis, Saarland University

Python, Computer Vision, OpenStreetMap, Google Street View, MaStR

- Developing a computer vision pipeline to detect balcony photovoltaic installations from street-level imagery .
- Built a geospatial pipeline using OpenStreetMap road networks for sampling and Google Street View for street-level image acquisition.
- Integrated MaStR, as an administrative reference and validation dataset for detected photovoltaic installations.
- Mapping detected installations to support local energy-grid capacity planning and analysis of distributed solar adoption.

HEALY

Saarland University

Python, Mistral, LangGraph, SHAP, RAG

- Developed a LLM agent using Mistral and LangGraph for explaining healthcare cost predictions from the SPARCS dataset.
- Integrated explainable machine learning using SHAP to provide interpretable explanations of model predictions and key cost drivers.
- Built a RAG-based evaluation pipeline using LLM-based G-Eval scoring, achieving approximately 0.95 precision on evaluation tasks.

LearnAid

Saarland University

React Native, FastAPI, LLMs, Recommendation Systems

- Led the development of a React Native AI tutoring application in a small team, delivering an end-to-end mobile learning experience.
- Integrated an LLM-powered feedback layer to provide contextual and personalized guidance based on learner performance.
- Developed a recommendation engine to generate personalized learning paths and suggest relevant educational content based on user progress.

Publications

Attention-based COVID-19 Detection using Generative Adversarial Network

2021

IEEE ICCIS 2021

GANs, Deep Learning, Attention, Computer Vision

- Co-authored research proposing an attention-based deep learning approach for COVID-19 detection from chest X-ray images.
- Applied Generative Adversarial Networks (GANs) to generate synthetic COVID-19 images and improve learned representations for COVID-19 and normal chest X-rays.
- Evaluated the approach on public medical imaging datasets using deep learning architectures including MobileNet and ResNet.

Certifications

- **Deep Learning Specialization** – DeepLearning.AI / Coursera
- **The Complete Web Developer: Zero to Mastery** – Udemy
- **Neural Network from Scratch in TensorFlow** – Coursera
- **Introduction to Deep Learning & NLP in Python** – DataCamp

Languages

- **English:** C1 – Advanced
- **German:** A2 – Elementary